

Codex Hackathon

InSync Summit 2026



AI Workforce Planner

AI-assisted workforce planning that brings talent, availability, opportunity demand, and staffing evidence together for faster, better decisions.

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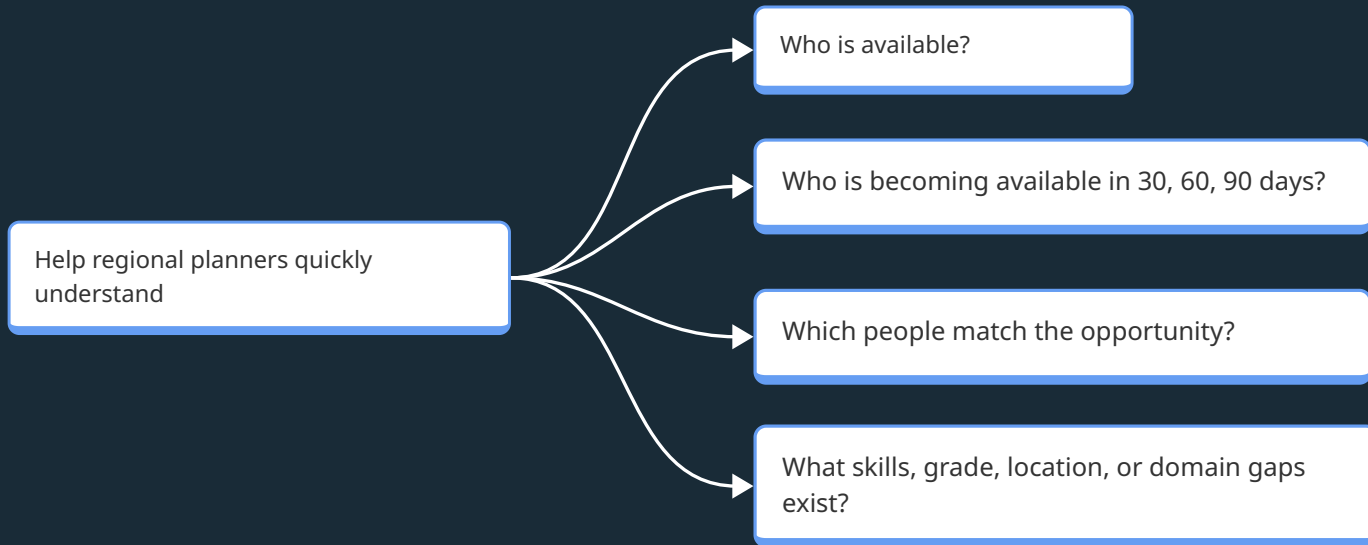


Tan Jian Quan

Software
Engineer Intern

The Challenge – Regional Challenge Tackle

- Build AI-powered workforce planning assistant



The Challenge – Specific Pain Point



The staffing evidence is scattered across spreadsheets, workforce data, skills records, project history, bench information, and individual knowledge.

Planners spend too much time collecting information before they can recommend a team.

The Challenge – Why This Matters Now



- 1 Opportunities can be delayed while planners search across systems.
- 2 Bench talent may stay invisible until too late.
- 3 External sourcing can be considered before internal options are exhausted.
- 4 Decisions depend on individual memory instead of shared evidence.
- 5 Sales and delivery need credible staffing options within minutes.



Solution in Action



1

Dashboard

Open workforce dashboard

2

Dashboard

Review 30 / 60 / 90-day availability

3

Opportunity Intake

Select or create an opportunity

4

Recommendations

Generate staffing recommendations

5

Recommendations

Compare three team options:

- Best Skill Fit, Fastest Available, Balanced Low Risk

6

Recommendations

Review AI explanation, risks, gaps, and next actions. Select a option

7

EWA

Prepare EWA review pack

8

Analysis

Visualize Skill Gap Analysis, Workforce Forecasting, Opportunity Forecasting

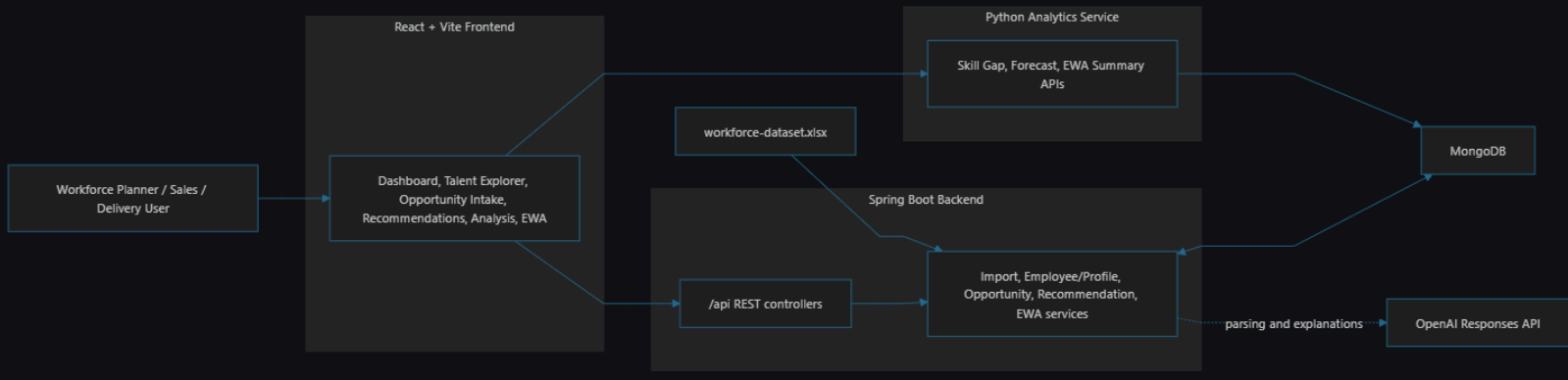
Working Prototype Link: [MusangKingDemoVideo.mp4](#)



Insync Summit 2026



Under the Hood – Architecture Diagram



Under the Hood – Codex Delivery



Build React pages
and improve UI
layouts

Debug Spring Boot,
MongoDB, Python
analytics, and
frontend integration
issues

Generate analysis
logic for skill gaps,
workforce
forecasting, and
opportunity demand



Design backend
entities, repositories,
and APIs

Refine README,
Docker setup, and
testing guidance

Under the Hood – AI Integration



Opportunity understanding

- Users can describe a staffing need in natural language, and the system converts it into structured opportunity and role requirements.

AI Workforce Planner

Dashboard

Talent Explorer

Opportunity Intake

Recommendations

Analysis

Opportunity Intake

Requirements parsed successfully. You may generate options now.

Opportunity statement *

need 3 roles in insurance domain, a business analyst, cloud engineer, and qa automation engineer. lead level is preferred. Required skills such as requirements gathering, aws, terraform, test automation, and selenium. stakeholder management, kubernetes, and performance testing are optional. each required 1 fte, except for business analyst, its fte is 0.5 and can combine candidates. Hybrid is allowed for cloud engineer. qa automation engineer must available when project start.

Opportunity brief *

Support an insurance client in modernising claims operations through clearer requirements, scalable cloud infrastructure, and stronger automated testing to improve delivery speed and service reliability.

Opportunity name *

Digital bank modernisation

Client name *

ABCDEFG Bank

Client type

Payment, Banking Services

Domain *

Banking

Region

APAC

Country *

Malaysia

City

Kuala Lumpur

Expected start date *

09/07/2026

Duration weeks *

Conversion Probability *

VALIDATED PREVIEW

Digital bank modernisation

ABCDEFG Bank - Banking

Generate options for recommender

Opportunity brief

Support an insurance client in modernising claims operations through clearer requirements, scalable cloud infrastructure, and stronger automated testing to improve delivery speed and service reliability.

Parsed Requirement

Roles

Business AnalystCloud EngineerQA Automation Engineer

Skills

requirements gatheringAWSterraformtest automationselenium

Stakeholder managementkubernetesperformance testing

Constraints

BankingMalaysia locationEET timezoneLead level preferred

Hybrid allowedMust available when project start

Under the Hood – AI Integration



Evidence-backed recommendation support

- The system uses AI-assisted reasoning and scoring logic to recommend suitable people, explain why they fit, highlight gaps, and suggest next actions.

 AI Workforce Planner

Dashboard

Talent Explorer

Opportunity Intake

Recommendations

Analysis

Recommendations

Evidence-backed staffing options generated from skill fit, availability, grade, location, domain, and project evidence.

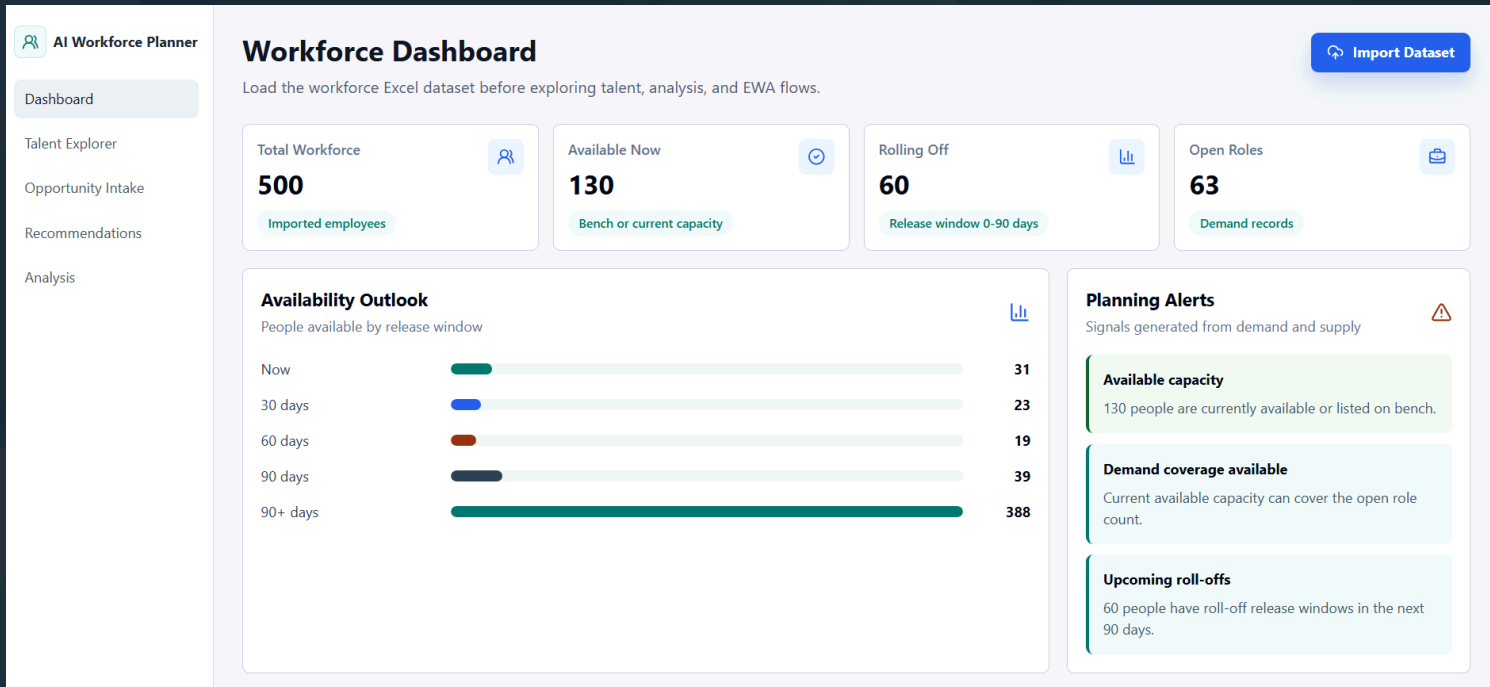
Recommendation run generated by the backend scoring engine.

	Claim Operations Modern... Banking / ABCDEFG Bank	 Start Date 9 Jul 2026	 Duration 13.8 months	 Location APAC, Malaysia, Ku...	 Key Roles 3	 Total FTE 2.5	
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 **AI explanation not generated yet**
Generate team and member explanations from this stored recommendation run. Backend scores and ranking will not change.

Generate AI Explanation

Result Screenshot – Dashboard Page



Result Screenshot – Talent Explorer Page

AI Workforce Planner

Dashboard

Talent Explorer

Opportunity Intake

Recommendations

Analysis

Talent Explorer

Search profiles by skill, role, grade, location, domain, and availability.

Availability: All

Region: All

Grade: Any

Domain: Any

Person	Role	Skills	Location	Availability	Domain	Fit	Fav
AN Aditi Nair Principal Consultant	Process Analyst	JIRA Stakeholder Management Workshop Facilitation	Chennai	Current Bench	Insurance, Public Sector	100%	
AM Aina Mustafa Associate Consultant	Batch Operations Specialist	On-prem Data Centre Operations Mainframe Operations COBOL	Penang	Current Bench	Legacy Platforms, Internal Platforms	100%	
AL Anh Le Lead Consultant	Frontend Engineer	AWS Java Spring Boot	Ho Chi Minh City	Current Bench	Banking, Media	100%	
AZ Aqilah Zainal Consultant	Mobile Engineer	Event-Driven Architecture AWS TypeScript	Kuala Lumpur	Current Bench	Payments, Travel	100%	
FR Farah Rahman Consultant	Design Engineer	Design Engineering Storybook React	Bengaluru	Current Bench	Banking, Travel	100%	
GA Grace Anderson Consultant	Solution Architect	Azure Cloud Architecture AWS	Sydney	Current Bench	Payments, Internal Platforms	100%	

Fav List

2

Save Filter

0

Result Screenshot – Person Profile Page

AI Workforce Planner

Dashboard

Talent Explorer

Opportunity Intake

Recommendations

Analysis

Talent Explorer

Person Profile

Dataset evidence used to understand this candidate.

AN

Aditi Nair

Principal Consultant | Process Analyst | Chennai

Current Bench

Remote

EWA: No Active Booking

CURRENT FTE

0

AVAILABLE FTE

1

PROJECT END

Jun 23, 2026

Availability

Calendar

Current Bench

Current project: No active project recorded. Planned end: Jun 23, 2026.

Primary Skills

JIRA

Stakeholder Management

Workshop Facilitation

Location And Domains

Chennai, India

Insurance

Public Sector

Profile Summary

Process Analyst based in Chennai with experience in Insurance and Public Sector. Core skills include Workshop Facilitation, JIRA, Stakeholder Management.

Project Evidence

Project	Client Domain	Role	Evidence	Dates
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Result Screenshot – Opportunity Intake Page

AI Workforce Planner

Dashboard

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Opportunity Intake

Requirements parsed successfully. You may generate options now.

Opportunity statement *

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Client name *

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Client type

Payment, Banking Services

Domain *

Banking

Region

APAC

Country *

Malaysia

City

Kuala Lumpur

Expected start date *

08/08/2026

Duration weeks *

55

Conversion Probability *

0.7

VALIDATED PREVIEW

Digital bank modernisation

ABCDEFG Bank - Banking

Generate options for recommender

Opportunity brief

Support an insurance client in modernising claims operations through clearer requirements, scalable cloud infrastructure, and stronger automated testing to improve delivery speed and service reliability.

Parsed Requirement

Roles

Business AnalystCloud EngineerQA Automation Engineer

Skills

requirements gatheringAWSterraformtest automationseleniumStakeholder managementkubernetesperformance testing

Constraints

BankingMalaysia locationGST timezoneLead level preferredHybrid allowedMust available when project start

Result Screenshot – Team Recommendation Page

AI Workforce Planner

Dashboard

Talent Explorer

Opportunity Intake

Recommendations

Analysis

Recommendations

Evidence-backed staffing options generated from skill fit, availability, grade, location, domain, and project evidence.

Recommendation run generated by the backend scoring engine.

Digital bank modernisation

Banking / ABCDEFG Bank

Start Date

8 Aug 2026

Duration

13.8 months

Location

APAC, Malaysia, Kuala ...

Key Roles

3

Total FTE

2.5

AI explanation not generated yet

Generate team and member explanations from this stored recommendation run. Backend scores and ranking will not change.

Generate AI Explanation

Option A: Best Skill Fit

SELECTED OPTION

81 / 100

CONFIDENCE

MEDIUM

RISK

Readiness

60 days

Members

3

FTE gap

1

Team Member	Availability	FTE	Gap	Score
AN Aditi Nair Business Analyst	10 Aug 2026	1	0	75
RA Ruby Anderson Cloud Engineer	10 Aug 2026	1	0	85
Anh Truong				

Option B: Fastest Available

LOW

100 / 100

CONFIDENCE

RISK

Readiness

2 days

Members

3

FTE gap

0

Team Member	Availability	FTE	Gap	Score
AN Aditi Nair Business Analyst	10 Aug 2026	1	0	75
RA Ruby Anderson Cloud Engineer	10 Aug 2026	1	0	85
MK Meera Kapoor	10 Aug 2026	1	0	74

Option C: Balanced Low Risk

LOW

77 / 100

CONFIDENCE

RISK

Readiness

2 days

Members

3

FTE gap

0

Team Member	Availability	FTE	Gap	Score
IK Ishaan Kapoor Business Analyst	10 Aug 2026	1	0	71
RA Ruby Anderson Cloud Engineer	10 Aug 2026	1	0	85
MK Meera Kapoor	10 Aug 2026	1	0	74

Result Screenshot – EWA Review Page

 AI Workforce Planner

Dashboard

Talent Explorer

Opportunity Intake

Recommendations

Analysis

EWA Review Pack

Selected team handover for approval and booking

3 EWA request(s) submitted successfully.

Opportunity Summary

Project

**Digital bank
modernisation**

Client

ABCDEFGH Bank

Domain

Banking

Location

**Kuala Lumpur, Malaysia,
APAC**

Start - End date

2026-08-08 - 2027-08-27

Duration

55 weeks

Selected Team Option

Option

Best Skill Fit

Match score

76%

Confidence

80.67%

Risk level

MEDIUM

Team size

3

Review status

Submitted To Ewa

Generated 1 Jul 2026, 12:11 am

Regional planner notes

Add notes before submitting to EWA

Export PDF

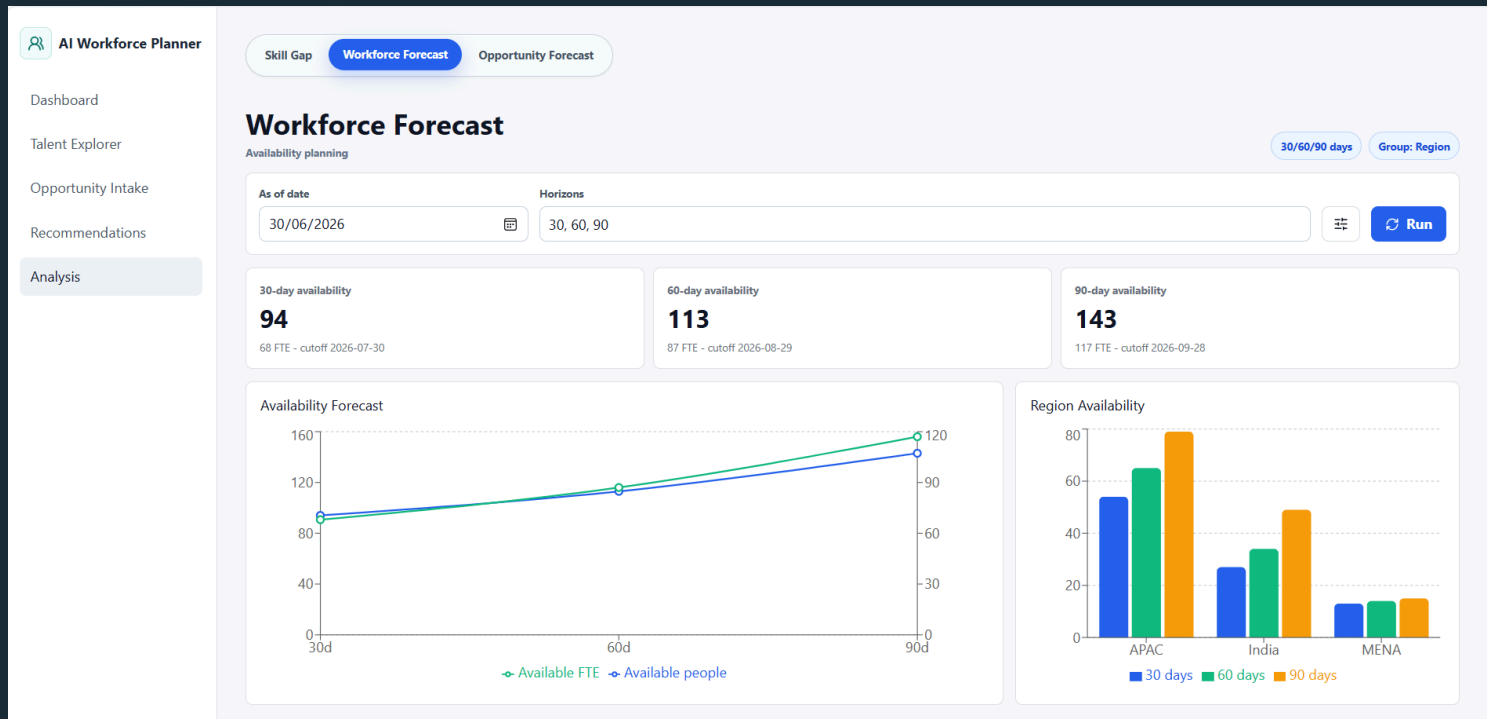
Submitted

Selected Overlay Candidates

3 people

Role	Employee	Scores	FTE	Skill Evidence
Business Analyst OPR-0064	Aditi Nair EMP-427	Overall 75% Capability 64% Availability 100%	1 available 0 FTE gap	Required 0/1 Desired 1/1
Cloud Engineer OPR-0065	Ruby Anderson EMP-054	Overall 85% Capability 79% Availability 100%	1 available 0 FTE gap	Required 1/2 Desired 1/1

Result Screenshot – Analysis Page





Under the Hood – Code Design

- **Modular service architecture**

React frontend, Spring Boot backend, Python analytics, and MongoDB are separated by responsibility and run consistently with Docker Compose.

- **Clean, maintainable backend design**

Controllers, services, repositories, DTOs, and entities are separated, making features easier to extend and debug.

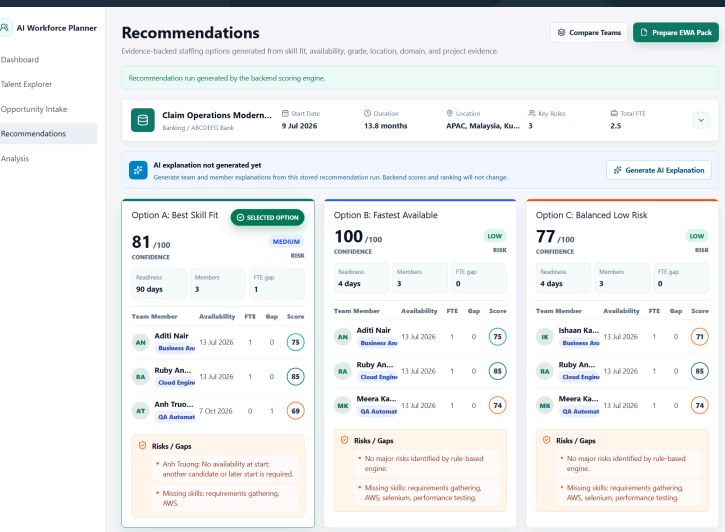
- **Auditable recommendation engine**

Rule-based scoring is transparent and explainable.

Stores generated RecommendationRun data so team comparison, AI explanations, and EWA flows can reuse the same output.



Business Value & Impact – Measurable Value



1

Staffing recommendation time reduced from hours or days to minutes

2

Higher internal utilisation by surfacing bench and soon-available people earlier

3

Reduced dependency on individual memory, spreadsheets, and informal conversations

4

Faster opportunity response, improving client confidence and proposal speed

5

Better workforce planning through 30, 60, and 90-day availability visibility



Business Value & Impact – How It Makes Endava More AI-Native

- This solution moves Endava from manual, person-dependent staffing decisions to an AI-assisted operating model where evidence is continuously available.
- Helping teams make decisions with explainable recommendations and confidence levels
- Turning fragmented workforce data into a searchable intelligence layer

AI Workforce Planner

[Dashboard](#)
[Talent Explorer](#)
[Opportunity Intake](#)
[Recommendations](#)
[Analysis](#)

Recommendations

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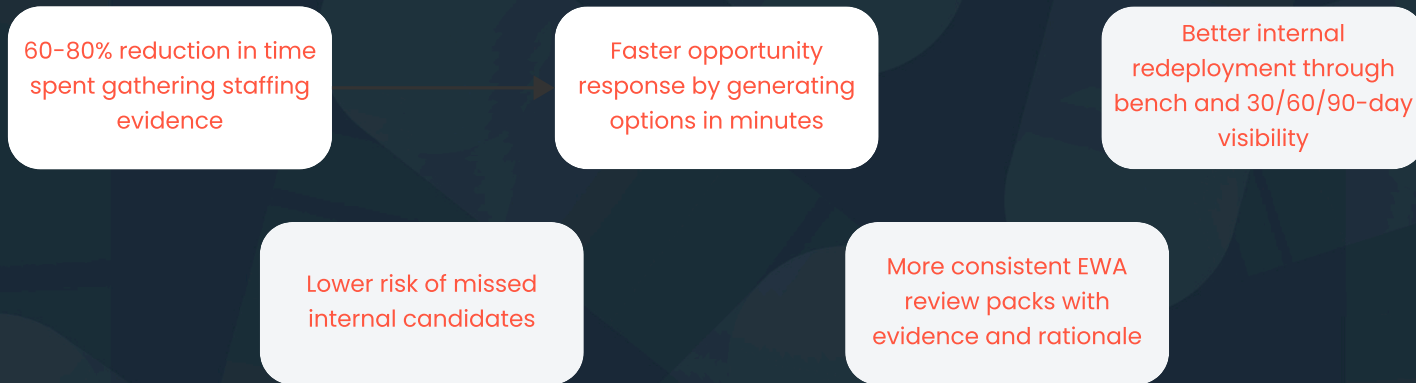
Generate AI Explanation

Compare Teams

Prepare EWA Pack



Business Value & Impact – Estimated Savings And Opportunities





Key Learnings & Next Steps – What we learned

What were your team's biggest technical or collaborative takeaways during the 4 days?

- Codex helped us turn broad challenge requirements into clear frontend, backend, data, and AI workstreams.
- We learned that workforce planning needs explainable evidence, not just a generated answer.
- Deterministic backend scoring is essential for trust; OpenAI is strongest when used to explain the score and next actions.
- Dataset understanding mattered as much as coding: people, skills, availability, project history, overlays, and EWA all had to connect.
- Human-in-the-loop design is critical: AI recommends, planners review, and EWA remains the final approval process.



Key Learnings & Next Steps – Next Steps

If you had another month to build, what features or reskilling pathways would you develop next?

- Conversational staffing assistant: ask questions like “Who can staff a banking AWS project in 30 days?”
- What-if scenario planning: simulate delayed start dates, extra roles, competing opportunities, or key people becoming unavailable.
- Skills gap and reskilling paths: show missing capabilities and recommend training or redeployment plans.
- Workforce forecasting: predict future supply, demand pressure, bench risk, and role shortages.
- Planner feedback loop: let users accept, reject, or adjust recommendations so future suggestions improve.
- Smarter EWA handoff: generate a complete approval pack with evidence, risks, FTE gaps, and next actions.

Thank you!

